

A Method of Estimating Residual Bending Error in GNSS-RO Absolute TEC

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ABSTRACT

Ionospheric refraction and dispersion effects induce residual bending error in the Total Electron Content (TEC) derived from Global Navigation Satellite System (GNSS) radio occultation (RO) measurements. This study proposes a practical method for the direct evaluation of residual bending error in the conventional absolute TEC retrieval. We derive the expression for the residual bending error in absolute TEC, by accounting for refraction residuals in both the carrier phase and pseudorange measurements. Two error components are identified: the altitudinally varying residual bending error from carrier phase TEC, and a constant bias from phase-to-pseudorange leveling. These errors are quantitatively assessed and analyzed by utilizing real observational data from COSMIC-2. Our analysis reveals that the residual bending error is predominantly influenced by vertical TEC gradients and can exceed the level of 10 TEC units (TECU). The findings underscore the need for estimating and correcting residual bending errors to improve the accuracy of absolute TEC retrievals, particularly under high solar activity conditions.

1 INTRODUCTION

The ionosphere has a significant impact on the propagation characteristics of radio waves. Ionospheric refractivity, which is a function of signal frequency and electron density, alters the propagation speed and direction of radio signals relative to vacuum propagation. This phenomenon induces group delay and phase advance in Global Navigation Satellite System (GNSS) signals traversing the ionosphere, which is approximately proportional to the integrated electron density along the ray path, known as the slant Total Electron Content (TEC). Due to the dispersive nature of the ionosphere, the ionospheric signal delay varies with signal frequency, enabling TEC estimation by differencing pseudorange or carrier phase (to be referred to as phase in the remaining paper) observables measured on two frequencies. The conventional dual-frequency method relies on the assumption of straight-line propagation, effectively eliminating non-dispersive terms such as the geometric distance and clock biases. However, ionospheric refraction causes signals at different frequencies to deviate from the straight-line path, leading to discrepancies between the estimated dual-frequency TEC and the true TEC along the straight-line. This discrepancy, referred to simply as the residual bending error, is particularly pronounced in the radio occultation (RO) geometry due to the extended ionospheric limb paths of signals traveling from GNSS to Low Earth Orbit (LEO) satellites. The residual bending error is mainly dependent on the vertical gradient of electron density, which varies with factors such as solar activity, local time, and the geographical location of the occultation event. Simulation studies have demonstrated that, under daytime conditions during solar maximum, the occultation ray path can deviate from the straight-line path by up to a few kilometers (Hajj & Romans, 1998; Hajj et al., 2000; Schreiner et al., 1999), leading to residual bending errors on the order of 10 TECU ($1 \text{ TECU} = 10^{16} \text{ electrons/m}^2$) (Høeg et al., 1998; Hoque & Jakowski, 2010). Hence, the residual bending error emerges as a significant error source for RO TEC estimation in the current solar maximum phase.

Over the past decades, various approaches have been developed to address residual bending errors in GNSS-RO measurements. Gu & Brunner (1990) and Brunner & Gu (1991) were among the earlier studies to introduce corrections for ionospheric refraction in the dual-frequency linear combination of phase paths, although their focus was primarily on ground-based GNSS positioning. However, this approach has practical limitations as it requires a priori knowledge of the refractivity gradients along the ray path. Subsequent studies by Høeg et al. (1998) and Syndergaard (1999, 2000) developed analytical models for the refraction residual in RO measurements based on spherical symmetry assumption. In such case, residual bending error is shown to be mainly dependent on the TEC gradient, which can be calculated directly from measurements. Nevertheless, the dependence on a priori knowledge is not completely resolved, since accurate estimation still requires auxiliary information on the electron density distribution. Similarly, Hoque & Jakowski (2011) proposed an empirical model based on ray tracing simulation results, which remains dependent on precise estimates for ionospheric parameters such as the maximum ionization. These limitations were improved upon by Syndergaard (2002) who proposed a model-independent method of combining dual-frequency ionospheric delay to eliminate first-order refraction effects. This approach leverages the frequency dependence of refraction residuals in phase observables, as derived by Syndergaard (2000), enabling the correction of residual bending errors using only dual-frequency phase measurements and precise orbit determination (POD) data. Following the validation of this methodology through ray tracing simulations in Syndergaard (2002), Chang et al. (2024) extended its application to real RO measurements from Constellation Observing System for Meteorology, Ionosphere, and Climate-2 (COSMIC-2), quantifying residual bending errors across various occultation events to assess the spatiotemporal variability of ionospheric refraction effects in the actual ionosphere.

In the present work, we propose a model-independent methodology for the estimation of residual bending error in GNSS-RO absolute TEC. Building on the work of Syndergaard (2000, 2002), which derived the residual bending error in relative TEC obtained from phase measurements only, this research incorporates pseudorange measurements into the analysis, which is needed to resolve phase ambiguities in relative TEC. This requires consideration of refraction and dispersion effects on both phase and pseudorange measurements, which propagate as additional errors in phase-to-pseudorange leveling for absolute TEC retrieval.

The outline of this paper is as follows. Section 2 provides a mathematical derivation of the residual bending error in absolute TEC based on the theoretical foundation from previous works. In Section 3, we present the proposed method for residual bending error estimation, including a brief overview of the data processing required for this methodology. Section 4 presents an illustrative example of the residual bending error analysis using COSMIC-2 data, with conclusions following in Section 5.

2 DERIVATION OF THE RESIDUAL BENDING ERROR

From the first-order expansion of the Appleton-Hartree formula for the ionospheric index of refraction (Papas, 1965), the phase and group refractive indices are written as follows (e.g. Bassiri & Hajj, 1993):

$$n_i = 1 - C \frac{N_e}{f_i^2} \quad (1)$$

$$n_{gr,i} = 1 + C \frac{N_e}{f_i^2} \quad (2)$$

where f_i is the carrier frequency, with subscript $i = 1, 2$ corresponding to two GNSS frequencies such as L1 (1575.42 MHz) and L2 (1227.60 MHz); N_e is the electron density, and $C = 40.3082 \text{ m}^3/\text{s}^2$. Higher-order terms in the expansion are beyond the scope of this paper, as their impact on residual error is relatively minimal compared to ionospheric refraction effects (Syndergaard, 1999). From Equations (1) and (2), the GNSS-RO phase and pseudorange observables in units of length are derived by the line integral of their respective refractive indices along the ray path between the GNSS and LEO satellites:

$$L_i = \int_i n_i ds \quad (3)$$

$$P_i = \int_i n_{gr,i} ds \quad (4)$$

where ds is the ray path element. Here, we restrict our region of interest to altitudes well above the neutral atmosphere, where the signal delay and bending are influenced only by the ionosphere. Hence, the neutral atmospheric refractive index is omitted from the equations above. Residual errors in phase and pseudorange observables due to ionospheric refraction may be separated by rewriting Equations (3) and (4) in a more convenient form:

$$L_i = \int_0 n_i ds + \left(\int_i n_i ds - \int_0 n_i ds \right) \quad (5)$$

$$P_i = \int_0 n_{gr,i} ds + \left(\int_i n_{gr,i} ds - \int_0 n_{gr,i} ds \right) \quad (6)$$

Here, the subscript $i = 0$ signifies that the integral is taken over a straight-line path. The residual terms in brackets are induced by ray path bending, namely the difference between curved and straight path integrals. Substituting phase and pseudorange refractive indices into Equations (5) and (6), and applying series expansions to estimate the refraction residuals to first order, the phase and pseudorange observables are expressed as:

$$L_i = \rho - \frac{C}{f_i^2} TEC_0 - \frac{1}{2} \frac{C^2}{f_i^4} \Gamma + B_i + \epsilon_i \quad (7)$$

$$P_i = \rho + \frac{C}{f_i^2} TEC_0 + \frac{3}{2} \frac{C^2}{f_i^4} \Gamma + B_{gr,i} + \epsilon_{gr,i} \quad (8)$$

ρ represents the direct geometrical distance between GNSS and LEO satellites, and the sum of all other non-dispersive effects such as clock biases and relativistic effects; TEC_0 is the straight-line TEC, namely the integrated electron density along the straight-line path; B_i is the phase ambiguity and bias; $B_{gr,i}$ is the code bias; ϵ_i and $\epsilon_{gr,i}$ are the sum of other error sources such as thermal noise and multipath for phase and pseudorange observables, respectively. Γ is known as the bending term, which arises as a result of ionospheric refraction. Generally, Γ is a complex function of electron density gradients along the signal path (Williams, 1975; Gu & Brunner, 1990; Melbourne et al., 1994). Under the assumption of a spherically symmetric electron density field, however, the expression for the bending term can be simplified as follows (Høeg et al., 1998; Syndergaard, 2000):

$$\Gamma = D \left(\frac{dTEC_0}{da} \right)^2 + a \frac{d}{da} \int_0 N_e^2 ds \quad (9)$$

Here, a is the straight-line tangent altitude, and D is the reduced distance, defined as

$$\frac{1}{D} = \frac{1}{\sqrt{r_L^2 - a^2}} + \frac{1}{\sqrt{r_G^2 - a^2}} \quad (10)$$

where r_L and r_G are radial distances to the LEO and GNSS orbits, respectively. The bending term is composed of two components: the major dispersion term, which is dependent on the square of the vertical gradient of TEC; and the minor dispersion term, which is dependent on the vertical gradient of the integral of squared electron density. As the name suggests, the major dispersion term is the dominant component of the total refraction residual (Syndergaard, 2000).

Straight-line TEC can be estimated from frequency-differenced phase and pseudorange observables, as shown below:

$$\begin{aligned} TEC_\phi &= \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} (L_1 - L_2) \\ &= TEC_0 + \Delta TEC_\phi^{\text{bend}} + \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} (B_1 - B_2) \end{aligned} \quad (11)$$

$$\begin{aligned}
TEC_\rho &= \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} (P_2 - P_1) \\
&= TEC_0 + \Delta TEC_\rho^{\text{bend}} - \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} DCB_{1-2} - \frac{f_1^2 f_2^2}{C(f_1^2 - f_2^2)} (\epsilon_{gr,1} - \epsilon_{gr,2})
\end{aligned} \tag{12}$$

$\Delta TEC_\phi^{\text{bend}}$ and $\Delta TEC_\rho^{\text{bend}}$ are residual bending errors in phase TEC (TEC_ϕ) and pseudorange TEC (TEC_ρ), respectively, and are written as follows:

$$\Delta TEC_\phi^{\text{bend}} = \frac{C}{2} \frac{f_1^2 + f_2^2}{f_1^2 f_2^2} \Gamma \tag{13}$$

$$\Delta TEC_\rho^{\text{bend}} = \frac{3C}{2} \frac{f_1^2 + f_2^2}{f_1^2 f_2^2} \Gamma \tag{14}$$

Note that the residual bending error in pseudorange TEC is three times larger than in phase TEC. Phase TEC has significantly lower noise levels and multipath errors (which were omitted from Equation (11)) compared to pseudorange TEC, but is biased by phase ambiguities. On the other hand, pseudorange TEC provides an absolute measure of TEC except for the offset from differential code bias (DCB) expressed as $DCB_{1-2} = B_{gr,1} - B_{gr,2}$. However, it is more susceptible to multipath and noise. Therefore, conventional TEC estimation involves the leveling of phase TEC to pseudorange TEC in order to remove the bias introduced by phase ambiguities (Mannucci et al., 1998):

$$TEC_\phi^{lvl} = TEC_\phi + TEC_\Delta \tag{15}$$

$$TEC_\Delta = \frac{\sum_k [(TEC_{\rho,k} - TEC_{\phi,k}) W_k]}{\sum_k W_k} \tag{16}$$

TEC_ϕ^{lvl} represents the leveled phase TEC, obtained by adjusting the level of the phase TEC (or ‘relative’ TEC) using the leveling constant TEC_Δ . The leveling constant is computed as the average difference between phase and pseudorange TECs, weighted by a signal quality-based factor W . The summation is performed across a phase-connected arc for which the phase ambiguity remains constant, and the subscript k refers to time in a given phase-connected arc. The leveled phase TEC still includes errors from DCB and multipath effects in the pseudorange TEC, but these errors are typically estimated and corrected separately to obtain the absolute TEC. The details of this error mitigation process are beyond the scope of this paper. Interested readers can refer to Morton et al. (2021). Therefore, the resulting expression for the absolute TEC is given as:

$$\widehat{TEC}_\phi = TEC_0 + \Delta \widehat{TEC}_\phi^{\text{bend}} \tag{17}$$

$$\Delta \widehat{TEC}_\phi^{\text{bend}} = \frac{C}{2} \frac{f_1^2 + f_2^2}{f_1^2 f_2^2} \left(\Gamma + 2 \frac{\sum_k (\Gamma_k W_k)}{\sum_k W_k} \right) \tag{18}$$

$\Delta \widehat{TEC}_\phi^{\text{bend}}$ is the residual bending error in the absolute TEC ($\widehat{TEC}_\phi$), which consists of two components: the phase residual bending error, as given in Equation (13), which originates from the bending effects in phase TEC; and the leveling residual bending error, which is a constant offset in the leveling constant (Equation (16)) resulting from the difference between the residual bending errors in phase and pseudorange TECs.

3 DATA AND METHODOLOGY FOR RESIDUAL BENDING ERROR ESTIMATION

As derived in the previous section, the estimation of residual bending error requires the knowledge of bending term Γ , which may be calculated using Equation (9), as proposed by Høeg et al. (1998). In practice, the unknown TEC_0 in the major dispersion term can

be replaced with the measurement-based TEC_ϕ . However, calculating the integral of squared electron density in the minor dispersion term requires a priori knowledge of the electron density distribution. While Abel inversion offers a method for estimating electron density from TEC measurements, this suffers from significant limitations of the spherical symmetry assumption. Particularly problematic are the results obtained for regions below the F-layer, where the assumption can introduce substantial systematic errors, even producing physically impossible negative electron densities (Høeg et al., 1998; Syndergaard, 2006).

Alternatively, if the POD information for the GNSS and LEO satellites are available, the bending term may be estimated using a purely measurement-based method (Syndergaard, 2002). The ionospheric delay in the phase measurement, referred to as the ionospheric excess phase, is defined as:

$$S_i \equiv L_i - \rho \quad (19)$$

The ionospheric excess phase S_i is derived by removing non-dispersive terms (represented by ρ), such as the geometric distance and clock biases, from the phase measurement L_i using precise orbit and clock information. Then, from Equation (7), the ionospheric excess phase is given as:

$$S_i = -\frac{C}{f_i^2} TEC_0 - \frac{1}{2} \frac{C^2}{f_i^4} \Gamma + B_i + \epsilon_i \quad (20)$$

Note that the ionospheric excess phase can only be determined up to a constant bias due to phase ambiguities. The frequency dependence of the refraction residual (proportional to f^{-4}) allows the ionosphere-free combination of dual-frequency excess phase measurements to eliminate the first-order ionospheric delay (proportional to f^{-2}) and isolate the bending term as follows:

$$\Gamma = \frac{2f_1^2 f_2^2}{C^2} \left(\frac{f_1^2 S_1 - f_2^2 S_2}{f_1^2 - f_2^2} \right) \quad (21)$$

Here, phase noise (negligible) and phase ambiguities are disregarded. In practice, we address the bias in the calculated bending term due to phase ambiguities by setting Γ to zero at positive elevation angles, neglecting the topside ionosphere and plasmasphere contribution to the refraction residual. These contributions may be non-negligible during high solar activity conditions and should be considered in future studies. Consequently, the residual bending error in absolute TEC can be calculated by substituting Equation (21) into Equation (18). To calculate the leveling residual bending error, we choose the weighting factor for the leveling constant to be the squared signal-to-noise ratio (SNR) (Yue et al., 2011; Pedatella et al., 2021). This method enables a reliable estimation of the residual bending error, calculated purely from real measurements without the use of ionospheric models.

Extracting the ionospheric excess phase from phase measurements requires careful estimation and correction of factors such as geometric distance, transmitter and receiver clock biases, and relativistic effects. This procedure is similar to the atmospheric excess phase processing required for the calculation of neutral atmospheric bending angles, as implemented in COSMIC Data Analysis and Archive Center (CDAAC) at the University Corporation for Atmospheric Research (UCAR) (Schreiner et al., 2010). Here, we highlight only the key aspects of the ionospheric excess phase processing algorithm implemented in this study, which is currently applicable to Global Positioning System (GPS) measurements only. A more detailed description of the algorithm is provided in Chang et al. (2024).

In this study, we employ a single-difference processing strategy that effectively eliminates LEO satellite clock bias by differencing data from the occulting GPS satellite with simultaneously sampled data from a reference GPS satellite (Schreiner et al., 2010). GPS satellite clock biases are corrected using a high-rate GPS clock solution at 5-second sampling, provided by Center of Orbit Determination in Europe (CODE) (Bock et al., 2009). Additionally, precise orbit data for LEO and GPS satellites are used to calculate the geometric distance, which can be obtained from CDAAC Level 1b POD files (leoOrb) and International GNSS Service (IGS) final GPS orbit ephemerides, respectively. Adjustments are needed to account for the fact that precise orbits are referenced to satellite mass centers while the true geometric distance is between the antenna phase centers (Schreiner et al., 2010). Hence, we implement an antenna offset correction for the LEO satellite by accounting for the vector from the mass center to the antenna reference point (ARP), using quaternions in the CDAAC Level 1a attitude files (leoAtt) to correct for satellite attitude variations. This correction is not currently applied to GPS satellites, as they maintain a relatively stable orientation during occultation events, producing negligible

errors (Hunt et al., 2018; Innerkofler et al., 2023). Corrections for the phase center offsets and variations have not yet been implemented.

4 RESULTS AND DISCUSSION

In this section, we demonstrate an illustrative example of residual bending error assessment using GNSS-RO measurements collected from the POD antenna of COSMIC-2. Figure 1 presents two selected occultation events that occurred on 19 June 2024, characterized by F10.7 solar flux index of 196, indicative of intense solar activity: Event 1, observed by Flight Model (FM) 5 for PRN11 setting at 09:27–09:48 UT (20:35–02:09 LT) (panels (a–c)); and Event 2, observed by FM 6 for PRN25 setting at 07:35–08:03 UT (12:14–19:38 LT) (panels (d–f)). The top two panels show the geographical location of the tangent points in white, superimposed on the vertical TEC map near the time of occultation, obtained from the IGS global ionosphere maps. The tangent points are defined for negative elevation observations in which the signal traverses the ionosphere below LEO satellite orbit altitude. The larger white dot on the tangent point trajectory indicates the location corresponding to where the residual bending error reached its maximum value. In the middle panels, the phase and pseudorange TEC are plotted as a function of elevation angle in blue and red, respectively. Note that the phase TECs are relative values, which were arbitrarily set to zero at maximum elevation angles. The bottom panels show the SNR of the L1 C/A signal as a function of elevation angle, which is used as the weighting factor for determining the leveling constant.

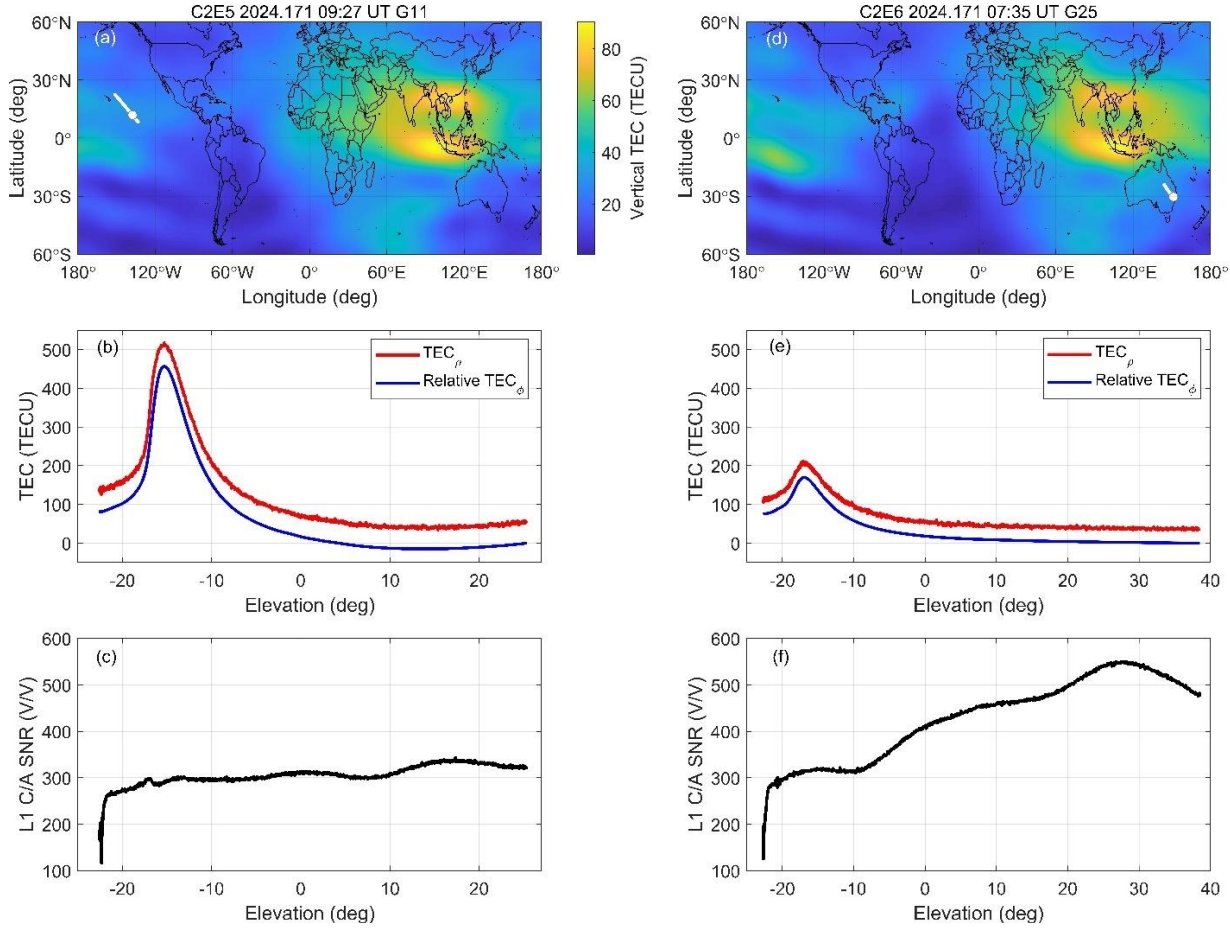


FIGURE 1 (a–c) Event 1 (COSMIC-2 FM5 PRN11 2024/06/19 09:27–09:48 UT). Top panel: Vertical TEC map and tangent point trajectory (white). Middle panel: Pseudorange (red) and phase (blue) TEC estimates. Bottom panel: SNR of GPS L1 C/A signal. (d–f) Similar to (a–c), but for Event 2 (COSMIC-2 FM6 PRN25 2024/06/19 07:35–08:03 UT)

Figure 2 illustrates how the calculation of the leveling constant is affected by the bending error in phase and pseudorange TECs. The top panels show the difference between phase and pseudorange TECs in black, where the rapid fluctuation arises from pseudorange noise and multipath effects. The red dashed line is the leveling constant TEC_{Δ} derived from Equation (16), using SNR as the weighting factor. The bottom panels display the phase residual bending error as expressed in Equation (13), computed using the proposed estimation method and set to zero at maximum elevation angles. Notably, the residual bending error in the topside ionosphere is negligible, consistent with its relatively low electron density gradient. However, upon comparing the negative elevation portion, panel (b) exhibits a large error peak, while panel (d) shows minimal variation in bending errors between the topside and bottomside ionosphere. Towards the end of the occultation event when the elevation angle drops below -20° , the bending error diverges due to neutral atmospheric signal delay. This non-dispersive delay is effectively canceled through frequency-differencing, and is therefore not visible in the top panels.

It is important to note that for Event 1, the sharp peaks in panels (a-b) align. The peak in the TEC difference relative to the background level of around 56 TECU ($92 - 56 = 36$ TECU) is approximately twice the magnitude of the peak in the phase residual bending error, which is around 17 TECU. This observation aligns with Equations (13) and (14), which indicate that pseudorange TEC contains residual bending error three times larger than that in phase TEC. Consequently, the sharp peak will have the effect of slightly overestimating the leveling constant and the subsequent leveled phase TEC. In contrast, Event 2 demonstrates minimal bending errors resulting in the TEC difference with a near-constant average, as shown in panels (c-d). In this case, the calculated leveling constant will primarily reflect the bias from phase ambiguities, assuming DCBs and multipath errors are mitigated, thus enabling a more accurate estimation of absolute TEC.

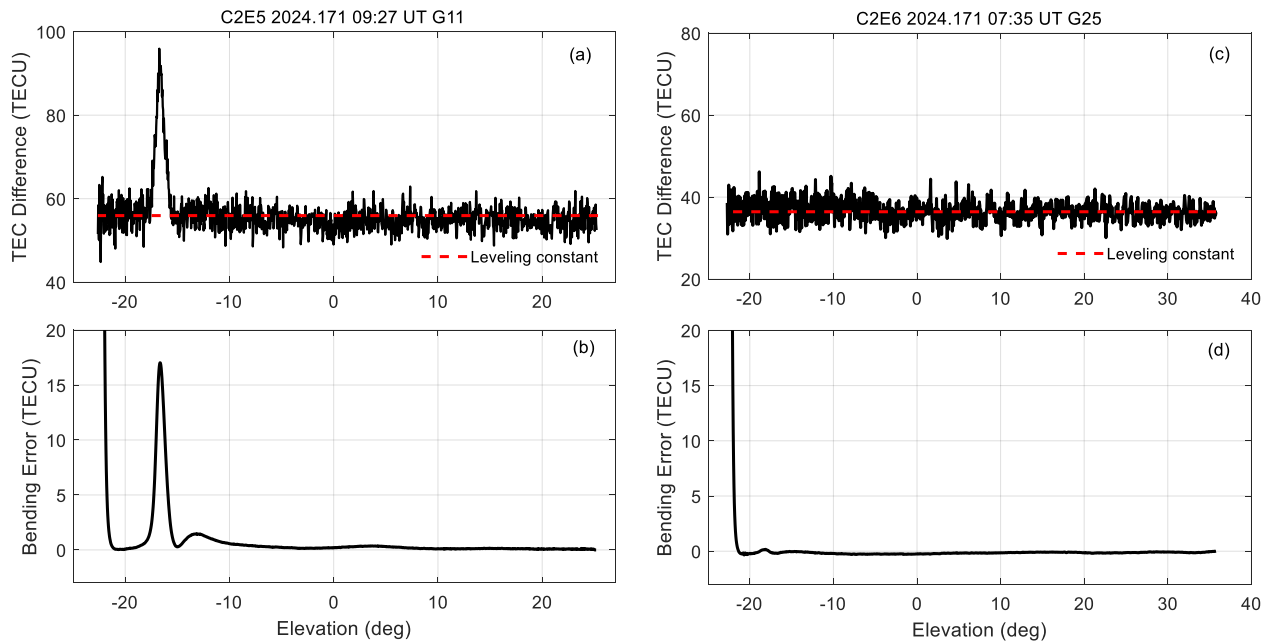


FIGURE 2 Relationship between phase-to-pseudorange leveling and residual bending errors. (a-b) Top panel: Difference between pseudorange and phase TECs (black solid line) and the estimated leveling constant (red dashed line) for Event 1. Bottom panel: Phase residual bending error for Event 1. (c-d) Similar to (a-b), but for Event 2.

In Figure 3, the top panels depict the pseudorange TEC (red) and the leveled phase TEC (black) as a function of altitude, taking only the negative elevation portion of the occultation events. The leveled phase TEC, obtained using the leveling constant from Figure 2, is an approximate value for the absolute TEC as it still contains transmitter and receiver DCBs as well as pseudorange multipath effects. However, we refer to this as absolute TEC for simplicity. The bottom panels display the residual bending error in absolute TEC as a function of altitude. The blue solid line represents the phase residual bending error, previously shown in Figure 2 as a function of elevation angle. For Event 2 (panel (d)), despite an overall minimal error of less than 0.5 TECU, the error peaks at the altitude of maximum vertical TEC gradient just below the F region peak, showing a similar pattern to Event 1 (panel (b)). The

dependency of the residual bending error on TEC gradients is corroborated by the similar trends between the blue solid and dashed lines, where the latter represents the residual bending error estimated by accounting for only the major dispersion term (i.e. squared vertical TEC gradient) in Equation (9). The green line indicates the leveling residual bending error, arising from the leveling constant computation shown in Figure 2. Since the leveling constant is calculated by averaging over the entire phase-connected arc, the effect of the residual bending error on phase-to-pseudorange leveling is somewhat diluted. However, this leveling error can still contribute a few TEC units to the total residual bending error, which is approximately 1.4 TECU for Event 1, and 0.24 TECU for Event 2. Finally, the black line is the total residual bending error in absolute TEC, calculated by the sum of blue and green solid lines. Note that all plots in Figure 2 are truncated at the altitude of 80 km to exclude the effect of the neutral atmosphere.

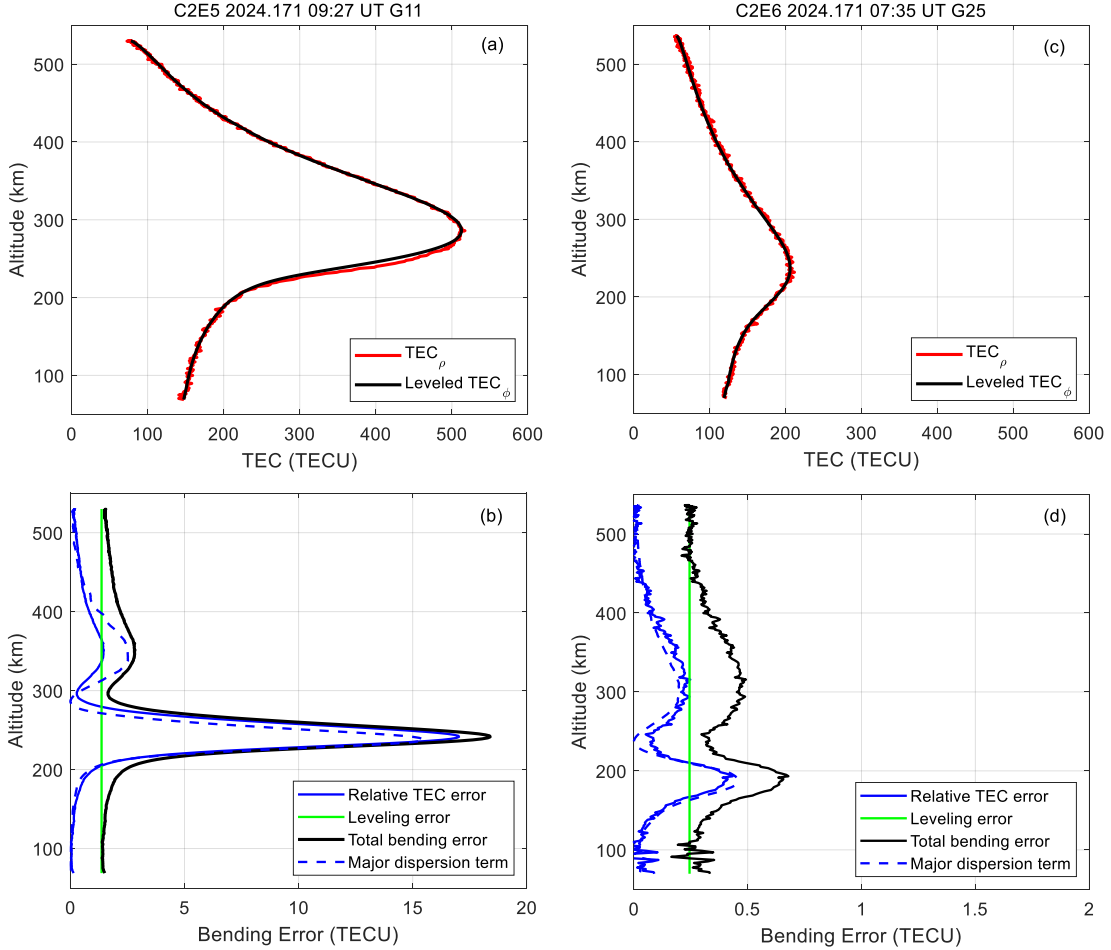


FIGURE 3 Residual bending error in absolute TEC. (a-b) Top panel: Pseudorange TEC (red) and leveled phase TEC (black) as a function of altitude, for Event 1. Bottom panel: Phase residual bending error (blue solid line), leveling residual bending error (green), and total residual bending error (black) as a function of altitude, for Event 1. The blue dashed line is the phase residual bending error calculated by accounting for only the major dispersion term in Equation (9). (c-d) Similar to (a-b), but for Event 2.

5 CONCLUSION

This study proposed a model-independent approach for quantifying residual bending errors in GNSS-RO absolute TEC retrievals, using dual-frequency ionospheric excess phase measurements. This methodology enables the estimation of residual bending error directly from measurements, without the need of a priori information on the electron density distribution. Using an illustrative example of an empirical analysis of residual bending errors using real occultation measurements from COSMIC-2, it was shown that residual bending errors can be a significant source of error under high solar activity conditions. Accounting for refraction residuals

in pseudorange measurements, mathematical expressions for two components of the absolute TEC residual bending error were derived: phase residual bending error, and leveling residual bending error. The leveling residual bending error, which is a constant bias arising from the phase-to-pseudorange leveling process due to the different refraction residuals in phase and pseudorange measurements, was shown to contribute a few TEC units to the total residual bending error. The altitudinally varying phase residual bending error typically has a more pronounced effect, especially in cases with high vertical TEC gradients where sharp bending error peaks are observed. It is possible to remove the leveling residual bending error by correcting both the phase and pseudorange TECs for their respective residual bending errors prior to leveling. Alternatively, it may also be effective to use only a portion of the phase-connected arc where the bending error is negligible, i.e. at positive elevation angles or high tangent altitudes (Hoeg et al., 1998). The remaining phase residual bending error can be estimated and corrected from the absolute TEC to enhance its accuracy.

As highlighted by Event 1 of our case study, large residual bending errors are often observed during post-sunset hours due to the presence of strong vertical electron density gradients in the ionosphere. Further statistical analysis is necessary to characterize residual bending errors across varying spatiotemporal conditions. This work establishes a foundation for such analyses, demonstrating the applicability of the proposed methodology for real ionospheric retrievals.

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